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Content Moderation in Social Media and AI

SUMMARY KEYWORDS

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Good morning, bonjour. I'm going to speak about content moderation in social media and AI. Let me share the screen now. So this is a topic of my presentation. First a few words to tell you where I'm coming from. I'm a computer science researcher at Inria French Government Institute. I'm also a board member at ARCEP, which is the French regulator of telecom, something like the French FCC. I'm also writing essays and novels. In the past, I've been a teacher in a number of places, including Stanford in the US. And I founded the startup Xyleme that's still existing. This is the organization of my talk, I will briefly talk about social media. But of course, you all know what this is. I'll talk about the responsibility of the social media, a little bit what's inside. And then we'll focus on content moderation, why it's difficult and why it's necessary to use machine learning. And then I'll conclude. The social media - it's important to realize how massive this is. There are 3.6 billion active users worldwide. Monthly, five social media are above 1 billion, and they're all from US or China. Facebook is very important in this setting. And not only for Facebook, but because of Instagram, Messenger, WhatsApp. And there are also other widely used social media that you probably know. It's important to see also that you find functionalities of social media in a number of digital services. I like Wikipedia, when you are an author in Wikipedia, when you're an editor, you are forming some kind of community with the other editors. And in a way, it's a social media that you're using. This is a technology that's really changing the world. For a number of aspects like enriching your relationships with other, tightening links with friends, meeting new people, you can express opinions, you can facilitate so-called democracy. But it also has lots of negative sides - addiction, hate speech, harassment, fake news, mass surveillance. You just open the newspaper, and you'll see lots of worries about these, these medias. So I'll argue now that it has a great power that we know, but also as such, it should have responsibility. So let's look at what is a state. And I think it's important to see this circle. I start with authority, a state as authority that's founded on legitimacy, TPP elections, and that's somebody is elected as a result of trust. And this trust comes from the fact that the state behaves responsibly in its application of authority. So really, you have this circle, that's essential to understand what a state is, and really this is something that you have to keep in mind when you think about social media. Now, state is a triptych, a territory, a population and authority. And if you think about it, a social media is also a triptych. A territory that's a portion of the cyberspace, a population, their users, we said, more than a billion sometimes. And authority and social media do have responsibility. For instance, they do some kind of coercion by closing accounts, deleting contents. So as such, they have greater and greater power and, somehow they behave like states. I don't know if you've heard of this idea of Facebook that to solve problem, they're going to introduce something that really looks like a Supreme Court and actually at the beginning, they were even calling it the Supreme Court.

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Now, once you find all this authority, you should ask the natural question - because you have authority, this comes from some kind of legitimacy. And now where is this legitimacy coming from for social media? Have they been elected? Certainly not. Are the huge profits bringing you some kind of legitimacy - that's doubtful. Maybe you give it to them by signing a contract. Yes, of course. But the contract is something like the terms of services that nobody reads. So you know, you cannot really be bound by something that you don't even know. The fact that you choose a particular social media that could be viewed as some kind of legitimacy, I mean, I chose them I cannot complain. Well, except that you cannot really choose, when all your friends on Facebook, what kind of choice do you have? You have to go there. If your friends move to Instagram, you move to Instagram, if they move to Snapchat, you move to Snapchat. So really, what is the choice? And same thing for Twitter and same thing for LinkedIn. So let's look at the origin of the problem. And for me, the origin of the problem in the US is section 230 of the Communications Decency Act in 1996. You may not have heard about that, but that's really essential. And actually, what's kind of funny, it's 1996, social media didn't even exist, so all the responsibilities come from something that lawyers designed without social media in mind, of course, they didn't exist. So what that's saying? It says that no provider or user of an interactive computer service shall be treated as the publisher, or the speaker of any information provided by another information content provider. What does that mean? It means that if you're a web platform, you certainly are not responsible for what others are publishing on it. So Facebook - you can publish anything you like, and Facebook is not responsible. And you can find similar problems with other things. Airbnb, for instance, if you look at section 230, if you're Airbnb, and somebody, some crooks publish ads for apartments on the platform, you're not responsible - this is their publication, they are in charge. So of course, this is not sustainable, and that's question more and more all over the world, even in the US. Now, is the problem only an American problem? No, because in Europe, we copied. And that's Article 14 of the e-commerce directive in 2001, which essentially says the same thing, essentially, not quite, there are fundamental differences between Europe and the US. And this comes from a fundamental difference in the interpretation of freedom of speech. Freedom of speech in the US is viewed as something religious, you cannot touch it. In Europe, if you look at the text, there are limitations of freedom of speech. And you can see that now with new laws that are coming from different countries, Germany, France, and others. In Germany, it's the NetzDG law something like the network Enhancement Act. And that is, that is something that is very recent, 2017, is something interesting, because this is an attempt to tame social media. And essentially, it's looking at one thing, it's hateful, disinformation and pornographic content. And there is an obligation, it says, okay, the law makes it that now, if there is a content that's illegal, you have to remove it within 24 hours, if it's clearly illegal. That is, if it's obviously illegal. And now as a lawyer, you're telling me how you define obviously, personally, I don't know. If it's just illegal, but it's not obviously illegal - you have seven days. We start touching the problem here, it's not very easy for the law to catch the notion of illegal content. And actually, the law didn't bring any. It didn't bring much, except for one thing - social media started understanding that governments were not satisfied and started discussing with the governments in Europe. And in Europe, there are laws coming - the Digital Service Act in particular, that should actually address the issue.

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Now, I know, I was complaining against Section 230 in the US and the Article 14 in Europe, but maybe they're right, maybe that's what we should do. And I believe that's not true. So let's look at the idea, the underlying idea. A social media platform is not a publisher. You know, social media does not select the content it publishes. So, a newspaper is responsible for the things it publishes. Why? Because the newspaper chooses what it publishes, social media does not. But there is a tricky aspect there, okay? Social media does not select what it publishes, but selects what it pushes. And, that's really a key idea because I mean, nobody cares if you wrote something stupid, something nasty, something hateful and it's read by five people. I mean, it's pretty much like you went to a bar, and you drank too much, and you said something that's outrageous, okay, that's bad. But it's not hurting the world, right? It's just a bad statement in a bar, okay? Now, the same thing is pushed to 200,000 people - now it's becoming an issue. It's becoming, it's seen as much as if it's published by a major newspaper. So that's a reason why I claim that social media platforms are responsible. Now, a couple of minutes to look at what it is from a computer science viewpoint. As I said, I'm a technical guy, I'm interested as well with what's inside. So let's open the hood. So I'm going to go from you to the data center, basically. So the user, there is an interface, which allows you to identify yourself, personalizes the content, there is some kind of encouragement to addiction. Typically, now you're using social media from smartphones, most people are using social media from the smartphone. Then there is network in charge of communication, the message is encrypted, compressed, that's very essential for this to work. And then at the other end, you have data center, and what's social media? At this stage, it's a database. It's really, it's a database, its content. Of course, there are lots of interesting functionalities, it's very important to..the pricing - all the most of the money comes from advertisements. So the algorithm for pricing the options is very essential for them. And we spend a lot of energy developing very smart techniques. And then there are two issues that are two functionalities. Critical recommendation, that's your traditional one, how do you choose among the hundreds of thousands, millions of tweets around which one is shown to you, that's essential, because basically, you're going to read, you know, dozens of tweets, not more. So the ones that the system decides not to show, you're never going to look at them. And moderation that's becoming more and more important. Before we get into that, let's look at something that's very, that's a fuel of everything. It's the personal data. And the data that these platforms acquire is really the value. And the data is first of all produced by users. When you type text, when you put a photo, and when you react, when you like a message, when you retweet one. So that's all things that you're directly producing as a user. And then there is all the knowledge that's inferred by the system by that analysis, machine learning more and more. For instance, what are the training products? For instance? How come that you and this guy that you've never seen in your life, like the same books, okay. Why do you care that you like the same books that this user that you never met, and you will never meet? Well, the system cares, because it's going to use this knowledge to push you books. And because of this personal data, the control of this personal data is becoming very important. That's why laws, like the GDPR in Europe, is really critical in this study. Recommendation, recommendation algorithms, that's a very active area of research in computer science. These social media are using very secret formula with large number of factors. The interest at the time, lots of things coming in.

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I just want to stress the fact that it's a critical, it's a critical formula with one example. In 2018, Facebook decided to give more importance to profile friends, family acquaintances for the newsfeed than the posts of companies and media. The Facebook stock dropped dramatically, that really shows that there is a close connection between this recommendation, the business value of the enterprise and also the interest of the users. If you have the wrong formula, basically people are going to move to some other social media. Now, I'm going to go into something that you're perhaps less aware of, but that's getting more and more important for social media, which is content moderation. There is one aspect of content moderation, that's a bit critical, which is recommendation, that's clearly something which brings money to the social media, because there's going to be a criteria for advertisement for the interest of users, for catching more users. Content moderation is less so, it's not revenue for the company, it's cost. It's interesting, basically, because if you don't do it properly, people are going to be pissed and they're going to move out perhaps. Or you may get, you know, problems with governments. So this is what we're talking about. And basically, I'm going to argue that machine learning is the right way of doing content moderation. So to see why we have to go and use something as nasty as machine learning, let's see first, why it's not easy. So I'm going to use the visa that you're probably familiar with, with big data, I'm going to reuse them. So first of all, the difficulty comes from the volume. The volume is billions of content published every day that, you know, potentially every single one of them has to be moderated. So how do companies do today? But some don't do anything. Vkontakte, in Russia, for instance, they decided they don't do moderation, okay? That easy, you don't do moderation, you just let the world get more and more rotten, that's, at least you don't have to invest in there. Some companies started using human moderators. For instance, Facebook, use tens of thousands of human moderators. Doesn't scale, because you have billions of content. Volume. Second one is variety. And we will see, we'll come back to that, you know, you have to block fake news, you have to block violence, you have to block hate speech, copyright violations - there are lots of things that you don't want to see on your social media. And velocity. Sometimes you have to react immediately. So let's take this example of the Christchurch terrorist attack in 2019. Again, two mosques in New Zealand, the guy who did that actually made a movie at the time he was doing it and the movie went basically on the platforms and could not be stopped. As a result, many countries and big tech companies signed an agreement. As far as I know, the US didn't sign but many developed countries did. And the idea was to work together to eliminate terrorist and violent extremist content online, and in particular, be able to do immediate reaction. Now, if you want to do immediate reaction, you have to do it electronically. You don't have the time to let a human moderator look at the video and decide that oh, this video is bad. It's not only one video because this video has perhaps been already replicated in five different sites. So you have to be super fast in reacting. That's a V of velocity. Now why it's not easy, it's not easy because of the ambiguity of human language. Humans speak with jargon, we slang with codes. In France, for instance, the Muslim was protected, you know, you could not say negative things about Muslims. So the extreme right people started talking about Islamist. Now, we all know that it's not the same thing, but in a way that's the way to go around the interdiction to say something harmful for Muslims. They would use the word Zionist for Jews and so on. So there is all this kind of coded language. And I showed you here, the most obvious one. But also, this is something that's changing all the time because they tried to react from against the platform that's blocking the account by using more and more code coded language. Figures of speech, humor, lack of context, who wrote this particular content, the same thing written by two different people as a totally different idea. So here, I have a book by Dany Lafferrière, when I was

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visiting the moderation center of Facebook, they were very happy to show that they detected this as racist content that they wanted to block. Except that, you know, this is not a racist content. This is actually the book that's written by an anti-racist guy, and he was actually using humour against racism. It has been translated for those of you who are interested. So, again, context, if you know that Dany Lafferrière wrote this book, and you know that Dany Lafferrière is a strong anti-racist advocate, then you know this content is not racist. Just to show another example of difficulty, the moderators of these platforms, typically for privacy reasons, they see only one content, sometimes if it's the answer to a message, they would see the original message. But basically they're completely lacking the total context. And that is really making it much more difficult to understand that, for instance, that this is harassment. Harassment is actually a good example. A study has shown that if you want to find harassment, you catch it better not by looking at the single content, but by looking at the message graph, forget the content, just look at the message graph, but by the look of the graph, you can see that it is very likely to be harassment. Moderation is not easy for humans. Open the newspapers look a bit. This is, you know, it's very difficult for a human being to sit in front of a screen and spend the days looking at hateful content. Because basically, to optimize their time, they don't let them look at random content on the internet, what they do is they say, well, this is a content that's likely to be hateful. Why? Because somebody notified it as hateful or because the program detected it as hateful. By the way, they're looking more and more to detection for programs because the detection of programs is actually getting better results than notification by users. User is basically very biased, and doesn't really know what to report. So lots of psychological issues in having people looking at porn all the time or looking at hateful content all the time, and so on. So if you cannot do it with humans, let's do it with algorithms, and what is the status now? It's reasonably efficient for terrorism and pedo pornography, which is where the algorithm got to a very high rate of success. It's much harder for hate speech, and even harder for fake news. Actually, sometimes fake news it's really in the eyes of the believer, I take an example - vaccine. I'm a strong believer in vaccine, but there are arguments that showed that sometimes some vaccines could be harmful. I can say that a particular vaccine is harmful without being a part of this global world craziness about saying that vaccines are bad for your health. Vaccines are saving millions of lives. They've probably saved, you know, hundreds of millions so far and they will continue and, for instance, with COVID and I'm really really, you know, looking forward to be vaccinated okay. But still, you know, particular message for vaccine can be totally correct, against vaccine can be totally correct. It's easier for text than images because understanding text we go much further. And very difficult to get information to get measures. It's not easy to measure, and also the people who really have the measure they don't want to make any kind of publicity about it. So I'll just give a private source because I spoke to a number of engineers. And actually, that's what I've heard a few times - it's algorithms are already better than humans for hate speech text, I studied more a speech. They're already better than humans for hate speech text. I don't know about fake news, but at least for hate speech it's better. Many systems algorithms are detecting much more content, bad content that humans actually at the beginning of COVID most of the moderation was done by algorithms basically, because, you know, the organization of moderators could not function. As I said, very difficult to have precise, publishable measures. And the status right now is you very rarely remove content based just on an algorithm, you ask a human to confirm the decision of the algorithm. Now, major difficulty in this context is a definition of what is an inappropriate content. Okay, so you have to come back to the origins. First of all, there are laws. The law says, you cannot publish this. And these vary from country

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to country, for instance, in France negation ism, by saying that she could not did not happen is forbidden or in Germany as well. But that's not true in the US, in the US you can write that if you want. So, laws vary from country to country, and now, platform does not have to stick to the law. Platform should remove everything that's unlawful, but the platform may decide to remove more material. So for instance, many platforms in the US decide that nudity should be avoided. There is nothing illegal in France, for publishing a picture of a nude person, but for most platforms, you cannot do that. Now, how do you do that? You specify that at the meta level, you've some kind of ontologies and rules that to define what's inappropriate. It's rather complicated in a content you have to say, well, what is the community that is attacked? And you know, is it to protect it, is this is a child or, you know, is this minority that's more protected? And then you have laws that say, well, this is harmful and this is attacking a minority that's protected. So this should not be allowed basically, you have some rules that you can see some form of applied ethics. Now, this is at the meta level and then you go to the data level. The data level is huge number of examples of content within a notation. Okay, this particular content is fake news, this particular content is hate speech, and so on. Now, once you get the new content, you can use a learning algorithm. So you've trained your algorithm, you got this large amount of content with a notation and then you say well, is this looking more like hate speech or not? And of course, you can do some smaller things, you know, you can try to guess what is the aggressor? What is the aggressive party, what the verb or what class over has been used and so on. And based on all these analyses, you can decide whether the content is appropriate or not.

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I'm getting to the conclusion. And of course, you have a situation that is pretty bad. Okay, people agree on one thing is that it's difficult to let it continue like that because it's hurting society. Now you have different ways to go. One way would be to say, well, let's forget about social media. Do we need social media? People have lived for centuries without so this was a bad invention, let's forget about it. Of course, we don't want to do that because this social media also bringing so much to us, as I said, people like it, okay. So one idea is to do regulation. And let's look at the regulation that have been considered so far. The first one is let social media self-regulate. And as I said, it doesn't work. It's criticized, heavily criticized, and the reason is fundamental it is if they moderate too much, they're violating the freedom of speech. If they moderate too little this brings harmful effects on their users. And the main reason is that they did not manage to build trust. And that's back to the original question I ask is, what is their legitimacy to do that? By defining what you can see and what you cannot see on these platforms, basically, they are shaping the society where we live, they're shaping the world of tomorrow. And why would they be allowed to do that? They're not legitimate. So there is another approach that have been chosen in countries that China and Russia and is - let the state do the regulation. I'm not sure we want that. So essentially, we believe that there is a different way to go. And that is gaining ground in Europe, which is, let's have the supervision of the social media by a regulator. So how does that work? The social media and the regulator specify goals together, reasonable goals, goals that would satisfy the society, and then the social media is totally transparent for the regulator. That means that the regulator can evaluate, can challenge, can criticize. As I said, it's very difficult to know whether the regulation works or not. I've spent months studying that, I've visited, I've discussed, we did particular work for Facebook and hate speech. I've discussed with Facebook, they opened up all their algorithm, their centers, we discussed with their engineers, and at the end of the day, I cannot really tell

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you, you know, what's the quality of their moderation. So, complete transparency so that the regulator can actually evaluate, challenge, criticize, and punish if necessary. Do you want to apply that to all the people that are doing social media? No, just to the big ones, to the huge one that really are shaping the society, structuring the society, they call it structuring platforms in Europe. And what's very important is that this should not just be a dialogue between the state and the platform, the entire society should participate, basically participate in defining the moderation they like. And by society, I mean here the associations, the academics, the justice system, and so on. And if you're interested, I give an article where we detail this, this approach. You can change the regulator, the regulation, but you can also change the business model. I didn't have time to talk too much about that, but if you look at it carefully, the business model is at the origin of everything, you know, the fact that we're getting these social media for free and that the companies that are offering them, they must make profit and they want to make big profits. So there are different approaches that have been considered, which I think are very promising. The first one is based on interoperability. You should look at telecom, okay, you can call with your telephone, another telephone that's perhaps with a different operator. I mean, it's not like you're given a telephone and you're told you can call only people with the same operator like you, you can call everybody okay? And you find that normal, you find that standard, okay. Nobody will see anything bizarre in the fact that you can call someone from another telecom operator. Now, when you move that to social media, it starts being totally different. Okay.

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I am on Facebook, you're on TikTok. If I want to, you know exchange content with them, I have to open an account on TikTok or you have to open an account on Facebook. Why is that? Is this because of technical reasons? Mastodon proves that it's not a technical reason, it's a social media that's based on open source software, you can build different instances of Mastodon and they can interoperate. And that's avoiding the network effects. Another approach that's, that's fantastic. It's WT.Social that was started in 2019 by Jimmy Wales. The business model is based on donations, no advertisement. And the idea is that the big value is not dollar or euro, the big value is truth. We want everything that's published to be truthful. And how do you maintain that by moderation, but the big thing is, who is doing the moderation - the community is doing the moderation itself. Okay. And now, we already mentioned an example of community doing its own moderation. It's Wikipedia, the editors in Wikipedia, they self organize. I'm not saying it's easy, sometimes it can be violent, okay, you can be yelling to, you can be yelled at, you can be yelling at people. But eventually, this settles down. And because you're a human being and you agree to speak and discuss, the idea that you can get to a solution. Last but not least, maybe we should change the users and education, education, education. For me, in the long run, this is the solution. You have to learn to analyze what you consume, which is if you're on Twitter, and you read something, then you should know how to read it, know that it's not because it's written on your board, and somebody wrote it, that it has to be true. Okay. It's kind of interesting that the people are doubting more when they read the newspapers, like the New York Times than what they read on Twitter. Come on, okay. You have to analyze what you consume. And also, when you publish something, you have to be careful, you have to think about what you're doing. When you're retweeting an information you have to think about what you do. So all this is possible, because users are also better informed. And as I said, the example of participation is Wikipedia. And I wanted to also mention something that I find very interesting is Moonshot CVE, the idea here is that if you detect that

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somebody published something that's inappropriate, okay, the standard thing to say is, oh, let's block that person. Now you block the person, is it solving the problem? No, you just block the person. Actually, in France, all the extreme right that was blocked on Facebook moved to Vkontakte in Russia. Actually, they don't bother the Facebook people, but they basically continue with their evil discussions somewhere else. Okay. So Moonshot CVE, the idea is that when you see someone publishing something inappropriate, maybe you block it that's not relevant, what you do is you talk to the person, try to convince the person why it's wrong, why the hate is not going to solve the problem, why this information that the person published is false, okay, that the earth is flat, I'll talk to you, I'll try to convince you that the earth is not flat, so that you won't build your own community of people thinking or claiming that the earth is flat. Okay, I'm getting to the end of this talk and I just want to do some advertisement. The Age of Algorithms is this, if you read English is probably what you want to get. The book is published originally in France and also translated in Chinese. If you read French, I've also recently written some short stories about the digital world. And also for the French readers, the Blog binaire in the newspaper Le Monde, and that's the French New York Times where with some friends we're publishing text about all this. And I'm thanking you for your attention and I'm going to try to stop this